

LDSCI6211 Machine Learning and Data Mining II AE3

Assessment Details

Course Leader:	Waleed Iqbal
First, Second, or Third Sitting:	First
Issue Date:	12 January 2026
Assessment Type:	Written Assignment
Assessment Title:	Final Project/ Written Assignment
Restrictions on Time/Length:	7 Pages / 2,500
Assessment Weighting:	30%
Hand-in Deadline:	By 13:00 on 24 April 2026
Planned Feedback Deadline:	28 calendar days after hand-in deadline or last presentation date
File Format Accepted:	pdf, .ipynb, zip
Mode of Submission:	Online (Canvas)
Anonymous Submission:	Yes
Marking Scheme:	Linear

	Prohibited	Use of AI is prohibited on this assignment.
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In this Module, you will learn about a wide range of deep learning applications. Part of the learning will be online, during in-class lectures and when attempting assessments, but you will really experience hands-on work in your final project. For it you need to choose wisely a project that fits your interests. One that would be both motivating and technically challenging. Your project should be an application project. Pick an

application that interests you and explore how best to apply learning algorithms to solve it.

This project is summative and assesses your ability to design, implement, and evaluate a deep-learning pipeline within the syllabus scope. You will be marked on technical correctness, experimental design, analysis/communication, and project management/reproducibility.

You shall create an end-to-end deep learning pipeline using a publicly available dataset. Your dataset choice should be aligned with an application problem.

Implementation Constraints

- Platform: Google Colab (free tier only)
- Framework: PyTorch or TensorFlow/Keras
- Dataset: Must be publicly available and accessible via standard libraries or direct download
- Computational Limits: Work within Colab's session limits (~12 hours, limited GPU access)
- Code: All code must be well-documented, reproducible, and include clear markdown explanations. All code should run without assuming installation of any package by evaluator. In case any package is needed to be installed, use in-notebook pip commands. For example, if Google colab does not have **nltk** package, your notebook should have ***!pip install nltk*** command running in a cell.

Architecture Requirements

You must implement and compare at least two different approaches from the following categories. Some example comparisons are given in mentioned three categories. You can also choose any comparison of your choice.

Category 1: Classical vs Deep Learning

- Classical ML method (Logistic Regression, SVM) vs Neural Network (MLP)

Category 2: Different Neural Architectures

- Feedforward Networks (MLP) vs Recurrent Networks (RNN/LSTM/GRU)
- Autoencoders vs Variational Autoencoders
- RNN vs Transformer-based models

Category 3: Architecture Variations

- Different RNN types (vanilla RNN vs LSTM vs GRU)
- Various autoencoder configurations
- Different attention mechanisms or transformer variants

Project Examples:

Choose any dataset of your choice which is publically available and apply your choice of architectural comparison on that dataset. Some example datasets from different domains and potential comparisons are mentioned. You can also apply any different comparison for given datasets.

1. Natural Language Processing

A. Sentiment Analysis & Text Classification

- **Dataset:** IMDb Movie Reviews, Amazon Product Reviews, or Twitter Sentiment
- **Approaches:** Compare Logistic Regression (TF-IDF) vs LSTM vs Transformer

B. Text Generation & Language Modeling

- **Dataset:** Gutenberg texts, song lyrics, or poetry datasets
- **Approaches:** Character-level RNN vs word-level LSTM vs GPT-style transformer

C. Text Summarization

- **Dataset:** CNN/DailyMail, BBC News Summary, or scientific paper abstracts
- **Approaches:** Extractive (attention-based) vs abstractive (seq2seq) methods

2. Time Series Analysis & Forecasting

A. Stock Price/Financial Prediction

- **Dataset:** Yahoo Finance, Quandl financial data, or cryptocurrency prices
- **Approaches:** Linear regression vs LSTM vs GRU with attention

B. Weather/Climate Forecasting

- **Dataset:** NOAA weather data, climate datasets from Kaggle
- **Approaches:** Traditional time series (ARIMA) vs RNN vs CNN-LSTM hybrid

C. Energy Consumption Prediction

- **Dataset:** Household energy consumption, smart meter data
- **Approaches:** Classical regression vs sequence models vs autoencoder-decoder

3. Audio & Signal Processing

A. Music Genre Classification

- **Dataset:** GTZAN, Free Music Archive (FMA), or Million Song Dataset
- **Approaches:** Feature-based SVM vs CNN (on spectrograms) vs RNN (on sequences)

B. Speech/Audio Analysis

- **Dataset:** LibriSpeech, Common Voice, or environmental sound datasets
- **Approaches:** Classical ML on audio features vs deep networks on spectrograms/waveforms

4. Recommendation Systems

A. Movie/Content Recommendation

- **Dataset:** MovieLens, Netflix Prize data, or book ratings
- **Approaches:** Matrix factorization vs autoencoder vs deep collaborative filtering

B. Music/Playlist Recommendation

- **Dataset:** Last.fm, Spotify datasets, or Million Song Dataset
- **Approaches:** Content-based vs collaborative filtering vs hybrid deep models

5. Tabular Data & Structured Learning

A. Healthcare/Medical Prediction

- **Dataset:** Heart disease, diabetes, or cancer datasets from UCI/Kaggle
- **Approaches:** Logistic regression vs deep neural networks vs ensemble methods

B. Economic/Business Analytics

- **Dataset:** Customer churn, sales forecasting, or marketing campaign data
- **Approaches:** Traditional ML vs deep networks vs generative models for data augmentation

Assessment Deliverables:

1. Project Proposal (20%) (K3c, S1c, T4c)

Deliverable: 2-page PDF proposal **Requirements:**

- Problem definition and motivation
- Dataset description and accessibility verification
- Proposed approaches and comparison methodology
- Expected challenges and mitigation strategies (if any)

2. Implementation & Technical Report (80%) (K1c, K2c, S2c, T3c)

Deliverable: Jupyter notebook + 7 page technical report

Technical Implementation (30%):

- Clean, well-documented code with clear explanations
- Proper data preprocessing and handling
- Correct implementation of chosen architectures
- Appropriate use of evaluation metrics
- Evidence of systematic experimentation

Technical Report (50%):

- **Introduction & Literature Review:** Problem context and related work
- **Methodology:** Clear description of approaches, architectures, and experimental setup
- **Results & Analysis:** Comprehensive results presentation with proper visualizations
- **Discussion & Conclusions:** Critical analysis of findings, limitations, and future work

Component 1: Project Proposal (20%)

Distinction Criteria (70-100%)

- **Problem Definition:** Exceptionally clear, well-motivated problem with significant practical relevance and potential impact

- **Technical Approach:** Sophisticated comparison methodology with thoroughly justified architecture choices and innovative elements
- **Dataset Analysis:** Comprehensive investigation of dataset characteristics, potential challenges, biases, and advanced preprocessing strategies
- **Planning:** Detailed, realistic timeline with clear milestones, risk assessment, and contingency plans
- **Innovation:** Creative problem formulation, novel application of techniques, or original research questions
- **Literature Context:** Demonstrates awareness of current research and positions work appropriately
- **Writing:** Publication-quality presentation with exceptional clarity and professionalism

Merit Criteria (55-69%)

- **Problem Definition:** Well-defined problem with clear motivation and practical relevance
- **Technical Approach:** Sound comparison methodology with good justification for architecture choices
- **Dataset Analysis:** Thorough dataset description with consideration of preprocessing needs and potential issues
- **Planning:** Realistic timeline with appropriate milestones and some risk consideration
- **Scope:** Well-balanced project scope showing good understanding of constraints
- **Research Awareness:** Shows familiarity with relevant techniques and applications
- **Writing:** Professional, well-organized presentation with minor issues

Pass Criteria (40-54%)

- **Problem Definition:** Adequate problem description with basic motivation
- **Technical Approach:** Reasonable approach selection with basic justification for comparisons
- **Dataset Analysis:** Basic dataset description with acknowledgment of main requirements
- **Planning:** General timeline with major milestones identified
- **Scope:** Appropriate project scope for available time and resources
- **Writing:** Clear and organized, with minor presentation issues

Fail Criteria (0-39%)

- **Problem Definition:** Vague, inappropriate, or trivial problem selection
- **Technical Approach:** Unclear methodology or inappropriate technique selection
- **Dataset Issues:** Inaccessible datasets or failure to verify data availability

- **Planning:** Unrealistic timeline or insufficient detail for execution
- **Scope:** Overly ambitious or trivially simple project scope
- **Writing:** Poor organization, unclear communication, or significant errors

Component 2: Implementation & Technical Report (80%)

Distinction Criteria (70-100%)

Technical Implementation (30%)

- **Code Quality:** Exceptionally clean, modular code with comprehensive documentation and professional software practices
- **Architecture Implementation:** Sophisticated implementation of multiple architectures with deep understanding and potential novel modifications
- **Experimental Design:** Rigorous experimentation with proper controls, extensive hyperparameter optimization, ablation studies, and statistical validation
- **Data Handling:** Advanced preprocessing, data augmentation, and handling of edge cases with consideration for data quality and bias
- **Evaluation:** Comprehensive evaluation using multiple metrics, cross-validation, statistical significance testing, and error analysis
- **Efficiency:** Optimal resource utilization with evidence of performance optimization and memory management
- **Innovation:** Implementation goes beyond requirements with creative solutions or advanced techniques

Technical Report (50%)

- **Literature Review:** Comprehensive, critical review positioning work within current research landscape
- **Methodology:** Crystal clear, reproducible description with theoretical justification for choices
- **Results:** Exceptional results presentation with insightful visualizations, thorough statistical analysis, and meaningful comparisons
- **Analysis:** Deep, critical analysis demonstrating sophisticated understanding of model behavior and failure modes
- **Discussion:** Thoughtful discussion of broader implications, limitations, and substantial future research directions
- **Originality:** Contains original insights, novel applications, or research-level contributions

- **Writing:** Publication-quality technical writing with perfect organization and professional presentation

Merit Criteria (55-69%)

Technical Implementation (30%)

- **Code Quality:** Well-structured, clean code with good documentation and clear organization
- **Architecture Implementation:** Correct implementation of required approaches with solid understanding demonstrated
- **Experimental Design:** Well-designed experiments with appropriate controls and systematic hyperparameter exploration
- **Data Handling:** Proper preprocessing with consideration for data quality and appropriate augmentation techniques
- **Evaluation:** Thorough evaluation using appropriate metrics with proper interpretation
- **Efficiency:** Good use of computational resources with some optimization considerations
- **Completeness:** All required components implemented correctly with some additional features

Technical Report (50%)

- **Literature Review:** Solid review of relevant work with good understanding and appropriate citations
- **Methodology:** Clear, detailed description of approaches with good theoretical grounding
- **Results:** Well-presented results with good visualizations and thorough analysis
- **Analysis:** Good analysis showing understanding of model performance and comparison insights
- **Discussion:** Meaningful discussion of limitations and realistic future work suggestions
- **Writing:** Well-organized, clear technical writing with good flow and presentation

Pass Criteria (40-54%)

Technical Implementation (30%)

- **Code Quality:** Clean, readable code with adequate documentation and comments
- **Architecture Implementation:** Correct implementation of at least two different approaches as required
- **Experimental Design:** Basic experimental setup with appropriate train/validation/test splits

- **Data Handling:** Proper data loading and basic preprocessing steps implemented correctly
- **Evaluation:** Appropriate metrics used with correct implementation and interpretation
- **Functionality:** All code runs without errors and produces reasonable results

Technical Report (50%)

- **Literature Review:** Adequate review of relevant work with basic understanding demonstrated
- **Methodology:** Clear description of approaches with sufficient implementation details
- **Results:** Complete results presentation with appropriate tables and figures
- **Analysis:** Basic analysis of results with reasonable explanations for performance differences
- **Discussion:** Acknowledgment of limitations and suggestions for improvement
- **Writing:** Clear, well-organized writing with minor issues

Fail Criteria (0-39%)

Technical Implementation (30%)

- **Code Quality:** Poor code structure, insufficient documentation, or significant errors
- **Architecture Implementation:** Incorrect implementation or failure to implement required comparisons
- **Experimental Issues:** Flawed experimental design, data leakage, or inappropriate evaluation
- **Data Problems:** Incorrect data handling or preprocessing that compromises results
- **Evaluation Errors:** Wrong metrics, incorrect implementation, or misinterpretation of results
- **Non-functionality:** Code that doesn't run or produces meaningless results

Technical Report (50%)

- **Literature Review:** Insufficient or inappropriate references, lack of understanding
- **Methodology:** Unclear or incorrect description of approaches
- **Results:** Incomplete, incorrect, or poorly presented results
- **Analysis:** Superficial or incorrect analysis, failure to understand results
- **Discussion:** No meaningful discussion of limitations or future work
- **Writing:** Poor organization, unclear communication, or significant errors

Submitting Assessments

For this assignment:

- Prepare a Jupyter Notebook file that contains the code and prepare a PDF file with the text and data visualisations, technical report.
- The notebook should be developed using Python 3 within a Jupyter environment, with well-commented code and appropriately structured text cells resembling a blog format.
- Remember to restart the kernel, thoroughly check the notebook for errors, and provide a working link to the dataset used in the analysis.
- Ensure that all code runs smoothly and create a compelling narrative by integrating analysis, visualisations, and reflections.
- Your submission may include auxiliary Python files (*.py), where you packaged, e.g., some common functions and imported them in your notebook. In this case, submit a .zip file with all documents included.
- Do not include any dataset files in your submission. If dataset is not loaded through repository, upload your dataset files on onedrive and share links of onedrive
- You have three submission attempts, but only the last submission will be graded. If your last submission attempt is late, you will receive the late penalty even if you have a previous submission that was on time. Please make sure to avoid multiple submissions for assessments with multiple components, as only the last attempt will be graded. Upload several files in one submission attempt instead.

If your assessment requires anonymous submission (see the assessment details table at the top of your assessment brief), please be sure you have left your name off of your submission and out of the submission file name, as failing to do so may result in a 0% mark on the assessment.

Refer to the assessment details table in your assignment brief for acceptable file formats. Avoid submitting zip files (unless explicitly required by the assessment brief); use the 'add files' function to submit multiple files instead. If you are submitting a physical artefact, you must also provide clear and thorough documentation (such as in the form of photographs or a video) of your submission by the deadline; see the bottom of this section for guidance on submitting video files.

Please ensure that you tick the agreement box at the very bottom of your Canvas submission page (scroll down if you don't see it). This will enable you to select 'Submit Assessment.' Please review the submitted file to ensure that everything is in order.

If you encounter any issues with submission, e-mail a copy of your assignment before the deadline to student.assessments@nulondon.ac.uk along with screenshots of the problem on Canvas, showing a timestamp.

To turn on notifications for submission confirmation emails in your Canvas settings: Account > Notifications > Turn on the bell for 'All submissions.' In the app this is via Settings > Email Notifications > All submissions.

To submit a video recording: Select the 'Panopto video' icon in the text entry box in your submission portal. You can upload a video file of any format from your media library by selecting 'upload,' choosing 'my folder' in the drop down menu, and clicking 'insert.' You should be able to play the video back once it processes. See further explanation, including guidance on recording videos using Panopto, in this support article: ['How to Submit a Video Assignment in Canvas.'](#)

Marking

The University uses two assessment marking schemes – one for undergraduate and one for postgraduate – to mark all taught programmes leading to an award of the University.

More detailed information on the assessment marking scheme and the criteria can be found in the Course Syllabus, available on the University's VLE.

Learning Outcomes

This assessment will enable students to demonstrate in full or in part the learning outcomes identified in the Course Descriptor.

On successful completion of this assessment, students should be able to:

Knowledge and Understanding

Knowledge and Understanding

K1c	Systematically understand machine learning techniques and fundamental concepts surrounding supervised, unsupervised and deep learning.
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| K2c | Design original software of varying levels of complexity for data pre-processing, models, training algorithms, and result visualisation. |
| K3c | Identify best-of-kind machine learning models and algorithms suitable for solving a practical problem associated with a data case-study. |

Subject-Specific Skills

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|-----|---|
| S1c | Critically evaluate the technical, social and management issues of building machine learning applications for a given data set. |
| S2c | Apply methods, tools and techniques at the forefront of the discipline to solve practical machine learning problems. |

Transferable Skills

- | | |
|-----|---|
| T3c | Display an advanced level of technical proficiency in written English and competence in applying scholarly terminology, so as to be able to apply skills in critical evaluation, analysis and judgement effectively in a diverse range of contexts. |
| T4c | Participate in the design and implementation of machine learning projects. |

Accessing Feedback

Students can expect to receive feedback on all summative coursework within 28 calendar days of the submission deadline or, if applicable, the last oral assessment date, whichever later. The 28 calendar day deadline does not apply to work submitted late. Feedback can be accessed through the assessment link on the Canvas course page.

Late Submissions

Please ensure that you submit your assignment well before the deadline to avoid any late penalties, as a submission made exactly on the deadline will be considered late. Please keep in mind that there may be differences between your computer's clock and

the server time, which can cause discrepancies, and that Canvas may take some time to process your submission.

Your Canvas submission portal displays two due dates: one is the deadline for your assignment, and the second is the latest possible date by which your assignment can be submitted late. Please make sure you submit by the assessment deadline in order to avoid late penalties.

If assessments are submitted late without approved Extenuating Circumstances, there are penalties:

- For assessment elements submitted up to one day late, any passing mark will receive 10 marks deducted or a threshold pass (40% for undergraduate students, 50% for postgraduate students), whichever is higher. Any mark below 40% for undergraduate students and below 50% for postgraduate students will stand.
- Students who do not submit their assessment within one day of the deadline, and have no approved Extenuating Circumstances, are deemed not to have submitted and to have failed that assessment element. The mark recorded will be 0%.
- For assessment subelements, late submission will result in non-submission penalties deducted according to the marking criteria above.

For further information, please refer to [AQF7 Part C in the Academic Handbook](#).

Extenuating Circumstances

The University's Extenuating Circumstances (ECs) procedure is in place if there are genuine circumstances that may prevent a student from submitting an assessment. If the EC application is successful, there will be no academic penalty for missing the published submission deadline.

Students are normally expected to apply for ECs in advance of the assessment deadline. Students may apply for consideration of ECs retrospectively if they can provide evidence that they could not have done so in advance of the deadline. All applications for ECs must be supported by independent evidence.

Successful EC applications for live oral assessments, including vivas, will result in a deferral of the oral to be organized by faculty, students, and Timetabling for a date as close as possible to the original presentation date. The deadline for supplementary materials, if assigned, will be carried forward by the length of the oral assessment extension.

Missing an oral assessment, including a compulsory viva, without an approved EC will result in a non-submission for the entire assessment and, accordingly, a recorded mark of 0%.

Students are reminded that the ECs procedure covers only short-term issues (within 21 days leading to the submission deadline) and that if they experience longer-term matters that impact on learning then they must contact [Student Support](#) for advice.

Under the Extenuating Circumstances Policy, students may defer an assessed element on only one occasion and may request an extension on a maximum of two occasions.

For further information, please refer to the [Extenuating Circumstances Policy](#) in the Academic Handbook.

Academic Misconduct

You must submit work for assessment purposes that is your own and meets good academic practice. Assessments must be completed strictly in accordance with the instructions outlined in the assessment brief. This includes ensuring that your work is appropriately referenced.

It is important to understand if artificial intelligence (AI) is permitted to be used or not, and if yes, the conditions for its use. For clarity, the assessment brief expressly states if AI can be used, and the parameters of its permitted usage. You are not permitted to use AI (in any form) if the assessment brief does not expressly state that usage is permitted.

Any concerns with the academic integrity of a submission will be addressed in accordance with the [Academic Misconduct Policy](#). This policy gives details of the different types of misconduct and the steps the University may take when a concern arises. Action may result in academic penalties being applied directly to you. You are advised to review this policy in full before completing and submitting any work for marking purposes. The University may also take steps to ascertain the authenticity of a submitted piece of work in cases of relevant concern, such as by holding a viva.

Version History

Title: Assessment Brief Template					
Approved by: The Quality Team					
Version number	Date approved	Date published	Owner	Location	Proposed next review date
2.0	December 2024	December 2024	Dr Alexandros Koliousis	VLE/ Faculty Resources Page	November 2027
1.1	July 2023	September 2024	Dr Alexandros Koliousis	VLE, Faculty Resources Page	November 2027
1.0	November 2022	January 2023	Dr Alexandros Koliousis	VLE	November 2027
Referenced documents	AQF7 Academic Regulations for Taught Awards; Extenuating Circumstances Policy; Academic Misconduct Policy; Course Syllabus				
External Reference Point(s)	UK Quality Code Theme: Assessment				

